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1. Which would reduce the corrosion resistance of stainless steel?

- a. Silicon
- b. Sulphur
- c. Manganese
- d. Magnesium

Sensitization; Vol 2, page 41, left, 1st; Vol 2, 3rd ed, 293, left

2. Increasing chromium content in a stainless steel _____.

- a. Increases heat resistance
- b. Decreases corrosion resistance
- c. Increases corrosion resistance
- d. Increases electrical conductivity

3. Adding carbon to stainless steel will _____

- a. Increase toughness
- b. Reduce corrosion resistance
- c. Increase temperature resistance
- d. Reduce fatigue strength

4. The frequency of the ultrasonic vibration is determined by _____.

- a. Pulser
- b. Frequency setting on filter
- c. dB gain
- d. crystal thickness

5. An important property of lipophilic emulsifier is _____.

- a. Viscosity
- b. Density
- c. Specific gravity
- d. Color

6. What kind of penetrant test should be done on a critical magnesium casting?

- a. Fluorescent, post-emulsifiable
- b. Fluorescent, solvent removable
- c. Fluorescent water washable
- d. Visible dye water washable

7. Which emulsifier is most sensitive for surface discontinuities?

- a. Lipophilic
- b. Hydrophilic
- c. Fluorescent
- d. Visible

8. For magnetic particle testing, what current would be best for detection of subsurface discontinuities?

- a. AC
- b. DC
- c. HWDC
- d. 3 Phase FWRAC or FWDC

9. What kind of current is used with prods?

- a. AC
- b. DC
- c. HWDC
- d. FWDC

10. What is the worst field to leave in a part after magnetic particle testing?

- a. Multidirectional
- b. Circular
- c. Longitudinal
- d. AC

11. Increased viscosity of penetrant will cause _____.

- a. Higher penetration rate
- b. Higher evaporation rate
- c. More dirt left on part
- d. Decrease rate of penetration

Vol 2, 3rd ed, rt, 1st, 85

12. A penetrant test is applied to a smooth part for small cracks. What sensitivity should be used?

- a. Low
- b. Medium
- c. High
- d. Ultrahigh

13. MFL detects discontinuities by _____.

- a. Contrast between particles accumulated at discontinuity and background
- b. Change in impedance
- c. Permeability change
- d. Flux leakage

14. A 15 psi pressure vessel with access to both sides would best be tested with _____.

- a. Vacuum box bubble leak
- b. Helium mass spectrometer
- c. Pressure change test
- d. Radiography

ASNT Vol 1, page 302, rt, last

15. What method would be used to determine the heat treat condition of an alloy?

- a. Magnetic particle
- b. Radiography
- c. Eddy Current
- d. Remote field

16. When no additional acoustic vibration is detected until the previous maximum load is exceeded, this is the _____.

- a. Barkhausen Noise
- b. Kaiser Effect
- c. Characteristic X-ray
- d. Felicity Effect

17. Which is an indirect method of magnetization?

- a. Head shot
- b. Leaches
- c. Prods
- d. Yoke

18. What is equivalent to J1 at 15 inch?

- a. 3.5 point font at 15 inch, Snellen 20/22 (@15 in), OrthoRater #9, or Titmus #10
- b. 4.5 point font at 15 inch
- c. J2 at 15 inch
- d. J3 at 20 inch

CP-189 para 6.1.1 and ASNT Vol 10, pages 52-54; TNT Article Vol 8, No 4

19. What is equivalent to J2 at 20 inch?

- a. 3.5 point font at 20 inch or OrthoRater #9
- b. 4.5 point font at 20 inch or OrthoRater #8
- c. J1 at 15 inch or OrthoRater #8
- d. J3 at 15 inch

SNT-TC-1A para 8.2.1, ASNT Vol 10 pages 52-54, TNT Article Vol 8, No 4

20. Two sensors used with MFL?

- a. Absolute and differential
- b. UV and Visible
- c. Hall Effect and Coils
- d. Survey Meter and Pocket Dosimeter

3rd ed Hdbk Vol 8, 329, rt, 4th; 10 256-257

21. Which developer is not permitted with visible dye?

- a. Nonaqueous solvent suspendible
- b. Water suspendible
- c. Plastic film
- d. Dry powder

3rd ed Hdbk, Vol 2, 38, left, 3rd

22. During immersion inspection using a zero degree probe on steel _____.

- a. Less sound is reflected than transmitted
- b. More sound is reflected than transmitted
- c. No sound is transmitted
- d. Reflected and transmitted sound are equal

23. Which of the following could leave a part with radiation?

- a. Fluorescent penetration
- b. X-ray radiography
- c. Gamma ray radiography
- d. Neutron radiography

24. Metal supports in a casting are called?

- a. Chills
- b. Chaplets
- c. Cope
- d. Drag

N151

25. What is used to prevent shrinkage porosity in a casting?

- a. Risers
- b. Sand
- c. Wax
- d. Chaplets

N142

26. What type of casting process applies pressure to the metal entering the mold?

- a. Centrifugal
- b. Lost Wax
- c. Sand
- d. Die

27. What type of discontinuity in a casting would be due to splash?

- a. Cold shut
- b. Unfused chill
- c. Scab
- d. Cold shot

28. How many orders of magnitude difference is there between milli and Mega?

- a. 3
- b. 6
- c. 9
- d. 12

29. What joining process uses a filler metal that melts at a temperature below the base metal and is performed below 840 degrees F?

- a. Adhesive bonding
- b. Soldering
- c. Brazing
- d. Welding

30. What joining process uses a filler metal that melts at a temperature below the base metal and is performed above 840 degrees F?

- a. Adhesive bonding
- b. Soldering
- c. Brazing
- d. Welding

31. Where would a mass spectrometer be placed to detect a leak in vacuum chamber?

- a. Inside only when using the tracer probe technique;
- b. Inside
- c. Outside
- d. Inside or outside

ASNT Hdbk Vol 10, pages 156-57

32. Which would not have a digital display?

- a. 6 inch scale
- b. Micrometer
- c. Caliper

3rd ed Hdbk Vol 9, 131-32

33. Which material cannot be tested with magnetic particle and penetrant testing?

- a. Ferritic SS
- b. Austenitic SS
- c. Martensitic SS
- d. Nickel

3rd ed Vol 10, page 275, rt, 2; 277 rt, 1. All can be done with PT, but not MT

34. A broad indication in the center of the weld shown in a radiograph would be

- a. LOP
- b. Concave root or suck back
- c. Lack of fusion
- d. Crack

35. A part has been in service for 15 years is magnetic particle tested. An indication is seen on the inside surface when tested from the outside. This is likely
- a. Corrosion
 - b. Corrosion fatigue
 - c. HAZ
 - d. Cracks

36. When performing a visual test, the first thing to look at is ____.
- a. The customer requirements
 - b. The standard
 - c. General part examination
 - d. Paperwork

3rd ed Hnbk, Vol 9, page 7, rt

37. A mass spectrometer may be used in Leak testing to ____.
- a. Leakage, location, and leak rate
 - b. Location, and leak rate only
 - c. Determine location for all test configurations
 - d. Size holes

ASNT Vol 1, page 371, rt last; ASNT Vol 10 156-57

38. As frequency is increased, standard depth of penetration during eddy current testing:
- a. Decreases
 - b. Increases
 - c. Is not affected
 - d. Doubles

39. During remote field testing the sensing coil is usually placed _____

- a. Next to the excitation coil
- b. In the transition zone
- c. In the null zone
- d. 2 to 3 tube diameters from the excitation coil

ASNT 3rd ed Vol 10, page 247, rt last

40. The material extruded between dies and subsequently trimmed off upon completion of forging is referred to as _____.

- a. Flash
- b. Seam
- c. Parting line
- d. Cold shut

3rd ed ASNT Hdbk Vol 10, page 304, left, last

41. Why are artificial discontinuities better than natural discontinuities for standards?

- a. This is false
- b. Artificial flaws are more reproducible and can be placed subsurface
- c. They have less repeatability and their response is equal to natural discontinuities
- d. Reproduceability using identical specifications

ASNT 3rd ed Vol 10, page 245, left 2nd

42. What conditions must be in satisfied in the inspection booth during fluorescent penetrant testing?

- a. Allow sufficient time for eye adaptation to darkness
- b. Ambient light should be no less than 2 ft-c
- c. Visible light at the inspection surface should be a minimum of 100 ft-c or 1076 lx
- d. UVA light intensity should be less than 1000 uW/cm²

ASNT 3rd ed Vol 10, page; Vol 2 3rd ed, 236, left

43. What does low viscosity do to capillary action during penetrant dwell?

- a. Increase time needed for penetrant to enter a discontinuity
- b. Decrease time needed for penetrant to enter a discontinuity
- c. Has no effect on penetrant entry into a discontinuity
- d. Increases the rinse time required for background removal

3rd ed Hnbk, Vol 10, page 102, left, 3rd; Vol 2, 3rd ed 45

44. For penetrants, a term used to classify relative sensitivity is _____.

- a. Brightness
- b. Process sensitivity
- c. Method
- d. Sensitivity level

3rd ed Hdbk, Vol 10, page 101, left 2 and 3

45. Which method requires pre-rinse?

- a. Water washable
- b. Post-emulsifiable lipophilic
- c. Solvent removable
- d. Post-emulsifiable hydrophilic

3rd ed Hdbk Vol 2, page 37, rt, 2nd; page 43

46. What is the most correct statement about HWDC as compared to DC surge, a technique where a very high current is applied for a short period, then reduced without interrupting it to a much lower steady current value?

- a. HWDC provides deeper penetration and superior particle mobility
- b. DC surge provides greater mobility but less penetration compared to HWDC
- c. DC surge provides deeper penetration and superior particle mobility
- d. The two provide equal mobility and penetration

3rd ed Hdbk Vol 8, page 32, rt, last to 33, rt 1st

47. What type of inspection is most often used for completed soldered joints and can be used to ensure penetration when there is access to both ends of the joint?

- a. VT
- b. UT
- c. PT
- d. MT

3rd ed Hdbk, Vol 9 page 248

48. What type of inspection is used to ensure penetration of filler metal when there is access to only one side of the joint?

- a. VT
- b. UT
- c. PT
- d. MT

3rd ed Hdbk, Vol 9 page 248

49. What are advantages of VT testing?

- a. Speed, economy and versatility
- b. Line of sight, access and speed
- c. Speed, economy, and subsurface sensitivity
- d. Limited shapes, speed and versatility

3rd ed Hdbk, Vol 10, 48 rt; Vol 9, v, left last

50. Which statement is true regarding a creep test and a fatigue test?

- a. A creep test uses a cyclic load and fatigue uses a static load
- b. Creep uses a static load, while fatigue employs a cyclic tensile load
- c. Both use static loads, but creep is performed at elevated temperature
- d. Fatigue tests are performed using static loads at elevated temperature and creep is performed using a static load with cyclic temperature

N76-77

51. What are the constituents of a green sand mold?

- a. Sand, water and clay
- b. Sand, water and thermosetting resin
- c. Sand, water and thermoplastic resin
- d. Ceramic powder, water and wax

N148-49

52. In manufacture of steel pipe by rolling strip and then using high frequency resistance welding with an upsetting force, lack of fusion would most likely occur

- a. Transverse to the weld
- b. Parallel to the weld and at the root
- c. Parallel to the weld at the crown
- d. Longitudinally and at any depth.

Vol 9, page 194, rt, 3rd and 4th

53. Probe resolution of the ultrasonic probe is determined by

- a. Separation of the sound entry signal from a near surface flaw
- b. Separation between two reflectors placed close together in time
- c. Half the wavelength
- d. Two closely spaced echoes along a lateral plane

UT PTP 61

54. Magnetic particle testing detects discontinuities by detecting _____.

- a. Permeability
- b. Leakage field
- c. Changes in conductivity
- d. Changes in impedance

3rd ed Hdbk, Vol 8, page 8, left 1; page 12, left, 1st

55. Multiple frequencies are used in eddy current testing to _____.

- a. Increase depth of penetration allowing detection as well as location
- b. Improving sensitivity to permeability changes when crack detection and wall loss are of maximum importance
- c. Increase sensitivity to a desired variable when many undesired variable would otherwise affect eddy current response
- d. Multifrequency is not used with eddy current

3rd ed Hdbk, Vol 5, 354, rt, 6th; 367, left

56. Which of the statements below is true about calipers and micrometers?

- a. A micrometer uses the space between a spindle and anvil to make a measurement, while a caliper uses separation between a pair of jaws.
- b. A caliper uses the space between a spindle and anvil to make a measurement, while a micrometer uses separation between a pair of jaws
- c. A micrometer analog, while a caliper is digital
- d. Neither device may incorporate a Vernier to increase resolution

3rd ed Hdbk Vol 9, 130-31

57. According to SNT-TC-1A, what is required to reinstate a fired Level III?

- a. Nothing as long as the employer has retained the records, the certification did not expire, and reinstatement is within 6 months of the termination.
- b. Re-examination
- c. Retraining followed by re-examination
- d. Restoration of the ASNT NDT Level III certificate

SNT-TC-1A 13 and 14

58. According to SNT-TC-1A, an employee has all the training hours, but only 40 hours of experience. What is required for the employee to work in the method?
- The individual may perform the work independently
 - The individual may record results, but they must be confirmed and signed-off by a Level II
 - Yes, but The individual must work under the supervision of a certified Level II.
 - Yes, but the individual must work with a certified individual

SNT-TC-1A para 4.2

59. According to SNT-TC-1A, during the course of obtaining necessary experience a trainee must ____.
- Work with a certified individual
 - Work under the guidance of a Level III
 - Work under the guidance of a Level II minimum
 - Not work with certified individuals

SNT-TC-1A para 4.2

60. According to SNT-TC-1A, before receiving training in PAUT or TOFD, the individual must ____.
- Be a Level III in UT
 - Have a Level I certification in the method
 - Have a Level II Certification in UT
 - Have a Level II Certification in acoustic emission

SNT-TC-1A Table 6.3.1 A, Note: 10

61. According to CP-189, during a practical exam what defects must be found?
- All indications large enough to detected and documented by the NDT Level III prior to administration of the exam.
 - All discontinuities or conditions documented and specified by the NDT Level III prior to administration of the exam
 - The discontinuities expressed by the Level III as the examined is provided
 - All discontinuities or conditions specified by the NDT Level III during administration of the exam

CP-189, para 6.4.3.2

62. According to CP-189, what happens to a Level III's certification if an individual's ASNT NDT III certificate expires?

- a. Suspended pending renewal of ASNT NDT Level III certificate
- b. The certification shall be revoked and the individual must under re-examination
- c. The individual shall be re-examined
- d. Revoked pending renewal

CP-189, para 7.2.4 and 7.4.1

63. At what point could a LINAC begin removing neutrons from an atom thus making the atom radioactive?

- a. This is not possible
- b. At 1 MeV
- c. At 3 MeV
- d. Between 3 and 5 MeV
- e. At 8 MeV and above

Neutrons are emitted by neutron-photo effect occurring inside the head. Outside the head, neutrons cause isotopes to be formed. Most activation occurs in the accelerator components to a lesser extent objects in room (including walls, floor and ceiling) and then minimally in air.

64. When using a LINAC at 8 MeV and above how long should a radiographer wait to enter the exposure room?

- a. This is not need as the power is off.
- b. At any time during exposure
- c. 30 seconds
- d. 2-4 minutes

65. Carbon Steel is _____?

- a. Non ferritic, Martensitic
- b. An Alloy
- c. Pearlitic
- d. Non ferritic, face centered cubic

N78, 1st

66. To increase the sensitivity of a vacuum box test you can?

- a. Decrease vacuum pressure
- b. Use a less viscous detector gas
- c. Add a liquid indicator
- d. Pressurize the vacuum box

LT Hdbk Vol 1, 290, rt col, 2nd para; 304, 2nd

67. Halogens in penetrant testing need to be controlled for which metal?

- a. Titanium
- b. Aluminum
- c. Magnesium
- d. 400 series stainless steel

NDT Hdbk Vol 10, 102, rt (2); Vol 2, 170, left, 2nd

68. What affect does increased viscosity have on the capillary forces required for Penetrant testing?

- a. Increase
- b. Decrease
- c. No effect

Principles & Applications of Liquid Penetrant Testing, page 23, specifically stated: Vol 2, 45, rt, 245, left, top rt although may not have any effect on time to penetrate small flaws. Vol 10, 102, left, 3rd and 85, left, 2

69. Fluorescent particles respond to wavelength of

- a. 365 nm or frequency of 833 THz.
- b. 550 nm or frequency of 833 THz.
- c. 550 nm or frequency of 545 THz
- d. 770 nm or frequency of 389 THz

Vol 8, 227, 223;

70. What is the definition of a leakage measurement?

- a. Locating leaks
- b. The rate at which gases or fluids escape a container
- c. Contamination by radioactive substances
- d. Hole size

Vol 1, 13, rt, 1st

71. Closed die forgings may need to be trimmed to remove_____.

- a. Dross
- b. Flakes
- c. Flash
- d. Bursts

N171-172 (last/first); 304, left, last

72. Which of the following would not require use of a filler metal?

- a. SMAW
- b. GMAW
- c. Oxy Acetylene
- d. EBW

73. Depth of penetration decreases when _____.

- a. Frequency increases
- b. Frequency decreases
- c. Conductivity decreases
- d. Permeability decreases

74. If you have 234 hours in Vibrations and 1035 hours in radiography, how many hours in vibrations are needed to become a Level II? 420/800, 1680/2400, Max 380 and 720 other hours.

- a. 1866 in vibrations and 65 hours in other methods
- b. 1931 in vibrations
- c. Vibrations is not a method
- d. A or b

CP-189 Appendix A

75. Acoustic emission utilizes resonant as well as _____.

- a. Narrow banded probes.
- b. Probes operating at frequencies of 100 KHz to 25 MHz
- c. Probes operating above 1200 KHz
- d. Broad banded probes.

ASNT Hdbk Vol 10, page 419, left, para 2 and 5

76. According to SNT-TC-1A, the term NDT applies to inspection methods used for _____ applications.

- a. Material inspection
- b. Flaw detection
- c. Predictive maintenance
- d. All of the above

SNT-TC-1A, para 2.1.15 (2020); para 2.1.11 (2016)

77. What type of inspection is used for performing continuous or on-line monitoring of equipment, SPC to determine future maintenance, and with the goal of performing maintenance at a point in time when it is most cost-effective and before equipment performance degradation or failure?

- a. NDT for materials fabrication and product technology
- b. Qualification and Certification
- c. PdM
- d. Probability of Detection (POD)

CP-189 (2020), para 2.1.20; 2.1.18 (2016)

78. According to SNT-TC-1A, an NDT Basic Examination questions should cover understanding of 1) SNT-TC-1A, 2) materials, fabrication, and product technology, and 3) other appropriate NDT methods. How would a PdM Basic differ?

- a. There is no difference in the subject matter
- b. 30 questions instead of 20 are recommended for other NDT methods
- c. Item 2 is on machinery instead of materials, fabrication and product technology
- d. b and c

SNT-TC-1A, 8.7.1 and 2.1.11 (2016); 8.7.1 and 2.1.15 (2020)

79. Which would have differences between the Basic exam administered for NDT and PdM?

- a. There is no difference between the Basic exam requirements in CP-189 and SNT-TC-1A, except in the number of questions.
- b. SNT-TC-1A
- c. CP-189
- d. SNT-TC-1A exam application package

SNT-TC-1A para 8.7.1; CP-189 para 2.1.18, 2.2.1 and application

80. What is the minimum water tolerance requirement for Lipophilic (Method B) penetrant emulsifier? (

- a. There is no limit as water is a constituent rather than a contaminant
- b. 5%
- c. 10%
- d. 15%
- e. 20%

AMS 2644

81. What is the minimum penetrant tolerance requirement for Lipophilic (Method B) penetrant emulsifier?

- a. 5%
- b. 10%
- c. 15%
- d. 20%

AMS 2644

82. What tracer gas is used because it is inert and safe for the ozone? (

- a. Halogens
- b. Nitrogen
- c. Helium
- d. Oxygen

83. Which welding process uses a small inert molecule as a shielding gas? (

- a. GMAW
- b. GTAW
- c. FCAW
- d. Shielded Metal Arc Welding

84. If the SFD is 72 inch, the SOD is 36 inch, and the object is 14 inch across, how large will the image of the object be? $D_{\text{object}}/\text{SOD} = D_{\text{image}}/\text{SFD}$

- a. 0.39 inch
- b. 5.14 inch
- c. 7 inch
- d. 28 cm
- e. 28 inch

85. 500 degrees C is equivalent to _____.

- a. 900 F
- b. 932 F
- c. 278 F
- d. 273 K

86. 100 degrees F is equivalent to _____ degrees

- a. 55.6 C
- b. 122.4 C
- c. 329 K
- d. 37.8 C

87. What would be the geometric unsharpness, U_g , in a radiographic image if the focal spot (f) is 3 mm, the source to object distance is 18 inches, and the 2 inch thick part is resting on the film? $U_g = f \times f/\text{SOD}$.

- a. 0.333 m
- b. 0.33 mm
- c. 0.013 inch
- d. b and c
- e. a and c

88. What weld is best suited for radiographic testing?

- a. T-Joint
- b. Corner joint
- c. Butt joint
- d. Edge joint

89. What NDT method utilizes Cambridge, fillet, throat and hi-lo gages?

- a. Ultrasonic
- b. Radiography
- c. Neutron radiography
- d. Visual

Vol 10, 3rd ed, 90, Fig 62

90. What causes cracking during a quench?

- a. Recrystallization during equilibrium cooling
- b. Mechanical stresses
- c. Thermal stress at the transitions between thick and thin sections
- d. Residual stress caused by absorption of hydrogen in the heat affected zone

Vol 10, 3rd ed 83-84, 305

91. Which of the following is a characteristic of the Hall Effect probes used in MFL testing?

- a. They are extremely thick solid state probes
- b. They are extremely thin, usually around 100 μm
- c. They are relatively thick solid state probes
- d. They are similar in construction to Forster probes

Vol 10, 257-258, Element 0.03 X 0.015 X 0.002 inch or 750 x 375 x 50 μm

92. While looking at archived film from 5 years ago, you notice that the film is fading/degrading. What is the cause?

- a. Developer
- b. Stop bath
- c. Rinse
- d. Drying

Kodak pdf 112 and 136; residual fixer

93. Understanding the relationship between the indication of a test and the result is important to:

- a. Establish control limits based on how these measurable parameters affect a product
- b. Increase cost of inspection
- c. Reject good parts
- d. Ensure all discontinuities found are rejected

M&P Ch 13, page 366, 2nd to 4th

94. Distance amplitude correction is used with ultrasonic testing _____.

- a. To compensate for different reflective area to establish a response curve of area to amplitude
- b. To compensate for loss of sensitivity caused by attenuation from scattering and absorption as sound propagates further into a material
- c. As a disciplinary measure taken against technicians who miss flaws
- d. To adjust for poor horizontal linearity

Vol 10, 318, rt, 1st

95. A disadvantage of the _____ test method is that a crack cannot be found without applying a stress to the part.

- a. Leak
- b. Ultrasonic
- c. Neutron radiographic
- d. Acoustic emission

~Vol 10, 416 rt to 417 left and M&P 359

96. What type of radiographic collimator would be used for a large vessel where film is overlapped and exposed in one shot?

- a. Annular
- b. Divergent
- c. Straight
- d. Hemispherical

~RT PTP 33

97. During angle beam immersion testing, which of the following waves is lost first?

- a. Shear
- b. Longitudinal
- c. Surface
- d. Transverse

~Vol 10, 324, rt, last

98. Eddy Current testing often uses frequencies between:

- a. 0 and 60 Hz
- b. Between 100 Hz and 10 MHz
- c. Between 1 and 15 MHz
- d. As high as 300 MHz

M&P 12.8.1, page 346

99. Transducers used for Acoustic emission are:

- a. Laser generated
- b. Electromagnetic acoustic
- c. Narrow banded and EMAT
- d. Resonant or broadband

Vol 10, page 419, rt col

100. The initial waves generated by an acoustic emission source are:

- a. Broad (wide) banded
- b. Narrow banded
- c. Dead zone
- d. Fresnel

Vol 10, page 419, left col, para 5th

101. Pulsed thermography would be classified as _____ technique.

- a. An Active
- b. A Passive
- c. An Inactive
- d. A Long pulse

Vol 10, 191, left, 1st; M&P 358, 2nd

102. Two thermography techniques are common in industry, passive and active. Active thermography techniques are:

- a. Based on introducing heat to the test object with an external heat source and then monitoring over time.
- b. Used to look for changes to the resistance of electrical flow, which causes an increase in temperature
- c. Concerned with friction heating or valve leakage or blockage
- d. Primarily used to inspect insulation and buildings

M&P 12.12.2, page 358

Question	Answer
1	b
2	c
3	b
4	d
5	a
6	c
7	b
8	c
9	c
10	c
11	d
12	d
13	d
14	a
15	c
16	b
17	d
18	a
19	b
20	c
21	d
22	b
23	d
24	b
25	a
26	d
27	d
28	c
29	b
30	c
31	b

Question	Answer
32	a
33	b
34	b
35	b
36	a
37	a
38	a
39	d
40	a
41	d
42	a
43	b
44	d
45	d
46	a
47	a
48	b
49	a
50	b
51	a
52	d
53	b
54	b
55	c
56	a
57	a
58	d
59	a
60	c
61	b
62	a

Question	Answer
63	e
64	d
65	b
66	b
67	a
68	c
69	a
70	b
71	c
72	d
73	a
74	d
75	d
76	d
77	c
78	d
79	b
80	b
81	d
82	c
83	b
84	e
85	b
86	d
87	d
88	c
89	d
90	c
91	b
92	c
93	a

Question	Answer
94	b
95	d
96	a
97	b
98	b
99	d
100	a
101	a
102	a

11/14/11

Additional Practice Questions

Q 28. Orders of magnitude equals factors of 10.

$$\text{milli} = 10^{-3} = 0.001$$

$$\text{Mega} = 10^6 = 1,000,000.$$

$$.001 \text{ to } 1,000,000$$

$$.001,000,000 = \underline{\underline{9 \text{ places}}}$$

Q 38

$$\delta = 26 \sqrt{\frac{1}{1000}}$$

f is in denominator. If you divide by a larger number, you get a smaller number

$$\delta \approx \frac{1}{\sqrt{1000}}$$

Q ~~38~~ 74.

RT 840/1600 Required

Vibrations 2100/5200 Required

Vibrations requires 2100 in method and allows 1100 from other methods

Min Vibrations 2100 - 2324 = 1866 hours

Min Other 1100 - 1035 = 65 hours

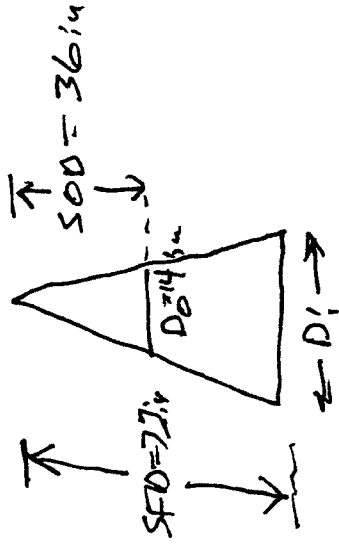
$$2324 + \underline{1866} = 2100 \text{ hrs vibrations}$$

$$\underline{65} + 1035 = 1100 \text{ hrs other methods}$$

$$\text{or } 1866 + 65 \text{ in vibrations} = \underline{\underline{1931 \text{ hrs}}}$$

Kraft Add. Honal Practice Questions

Q84.



Similar triangles:

$$\frac{b_1}{h_1} = \frac{b_2}{h_2}$$

$$\frac{D_0}{SOD} = \frac{D_I}{SFD}$$

$$\frac{14 \text{ in}}{36 \text{ in}} = \frac{D_I}{72 \text{ in}}$$

$$D_I = \frac{72 \text{ in} \times 14 \text{ in}}{36 \text{ in}} = \underline{\underline{28 \text{ in}}}$$

Q85.

$$1^\circ \text{C} = 1.8^\circ \text{F}$$

0°C is same as 32°F

100°C is same as 212°F

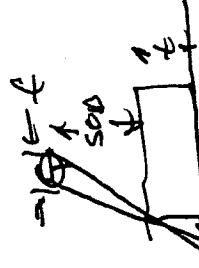
$$\frac{500^\circ \text{C}}{100^\circ \text{C}} \left| \frac{1.8^\circ \text{F}}{1^\circ \text{C}} \right| \frac{+32^\circ \text{F}}{+32^\circ \text{F}} = \underline{\underline{932^\circ \text{F}}}$$

Q86.

$$100^\circ \text{F} - 32^\circ = 68^\circ \text{F}$$

$$\frac{68^\circ \text{F}}{100^\circ \text{F}} \left| \frac{1^\circ \text{C}}{1.8^\circ \text{F}} \right| = \underline{\underline{37.78^\circ \text{C}}}$$

Q87.



Similar triangles.

$$\frac{U_0}{SOD} = \frac{U_1}{SFD}$$

$$U_0 = \frac{U_1 \times SOD}{SFD} = \frac{3 \text{ mm} \times 214}{18 \text{ in}}$$

$$U_0 = 25.4 \text{ mm}$$

$$\frac{.33 \text{ mm}}{25.4 \text{ mm}} \left| \frac{1 \text{ inch}}{25.4 \text{ mm}} \right| = \underline{\underline{0.013 \text{ inch}}}$$